of amino acids, with a indexing amino acid position and i the coordinate of the embedding space. The decoder takes the latent embeddings $z_{a,i}$ as input, and outputs corresponding token logits. The diffusion model is a conditioned transformer and we describe its action on the latent space below in the description of the diffusion loss. A more detailed description of the architecture is provided in Appendix A.

We train the auto-encoder under competition between a reconstruction loss and a normalization loss on the latent embeddings. For the reconstruction loss we employ the standard cross-entropy between the input tokens and output token logits. The normalization loss is less straightforward. In contrast to a variational auto-encoder Kingma & Welling (2014) which maps input to distributions over the latent space for which it learns the mean and variance, we let the encoder map directly to the latent space and guide the distribution of $z_{a,i}$ over a to be normally distributed (this foregoes the generative capability of the autoencoder, which is compensated for here by the diffusion model). Specifically, given a batch of sequences we employ a univariate parametric form of the Kullback–Leibler divergence

$$\mathcal{L}_N = \frac{1}{2d} \sum_{i=1}^{d} (\mu_i^2 + \sigma_i^2 - \log \sigma_i^2 - 1), \tag{1}$$

expressed in terms of the empirical mean μ_i and variance σ_i^2 of the $z_{a,i}$, with d the embedding dimension. We considered also a multivariate parametric form but found this to degrade performance, see ablation in Appendix C.

The simple combination of reconstruction loss and normalisation loss is not however sufficient to drive meaningful learning in the latent space. To achieve this, we consider two variants as follows:

- **Token Norm (LSD-TN):** here we modify the normalization loss by applying it separately to the embeddings of each amino acid type. Specifically, within each batch, we partition the latent embeddings into 20 sets, each corresponding to one of the 20 canonical amino acids, and employ a separate normalization loss for each set. While the default normalization loss allows different amino acid types to occupy distinct regions under the same normal distribution, this approach imposes a stricter constraint, creating an effective bottleneck for representation learning.
- **Noise Masking (LSD-NM):** here we modify the reconstruction loss to a variant of MLM designed for greater robustness to noise. Unlike standard MLM, where a fraction of amino acid embeddings are fully masked while the rest remain unaltered, our approach applies varying levels of corruption by inhomogeneously adding Gaussian noise to the latent embeddings. Specifically, we transform each amino acid embedding vector as

  $$z_a \to \cos(\pi t_a/2) z_a + \sin(\pi t_a/2)\varepsilon, \tag{2}$$

  where $t_a \in (0, 1)$ controls the noise level and ε is an embedding vector sampled from $\mathcal{N}(0, 1)$. To reflect this corruption in the reconstruction loss, we weight each embedding's contribution by the noise amplitude $\sin^2(\pi t_a/2)$, ensuring that highly corrupted embeddings dominate the training signal, while minimally corrupted ones contribute negligibly.

  We explored two sampling strategies for t_a: uniform sampling over (0,1) and sampling proportional to the signal amplitude $\cos^2(\pi t_a/2)$, as used for training the diffusion model (see below). The latter approach, which results in most amino acids being weakly noised while a few are strongly noised, performed better, and we adopt this choice in the models we present. See ablation in Appendix C.

  We emphasize that this inhomogeneous noising of the latent vectors is applied only during the training of the autoencoder. At inference the encoder maps deterministically to the latent space.

The diffusion model is trained on the autoencoder's latent space. For this we employ a variance-preserving cosine noise schedule Nichol & Dhariwal (2021), the v-target objective Salimans & Ho (2022), and epsilon prediction loss Ho et al. (2020), adopting the standard conventions of e.g. Hoogeboom et al. (2023). Specifically, the latent embeddings z get (here uniformly) noised to

$$z_t = \cos(\pi t/2) z + \sin(\pi t/2)\varepsilon, \tag{3}$$

for $t \in (0, 1)$ and $\varepsilon \in \mathcal{N}(0, 1)$, and the diffusion model is trained to learn

$$v_t = -\sin(\pi t/2) z + \cos(\pi t/2)\varepsilon. \tag{4}$$

Model	Encoder / Decoder	Diffusion
S	4.7M	7.3M
M	18.9M	29.0M

Table 1: Number of parameters for the S and M versions of the LSD model. The decoder is trivialised for the MLM diffusion baseline.

i.e. $\hat{v}_t = \texttt{Diffusion}(z_t, t)$. The epsilon prediction loss, expressed in terms of v, is weighted by the signal amplitude

$$\mathcal{L}_D = \frac{1}{2}\mathbb{E}_{t\sim(0,1),\, \varepsilon\sim\mathcal{N}(0,1)} \cos^2(\pi t/2)\|\hat{v}_t - v_t\|^2, \tag{5}$$

and we evaluate this with importance sampling. Indeed, from a representation learning perspective the increased weight for sampling t closer to 0 is intuitive, as information gets washed out with increasing t.

In this work we focus on the discriminative capability of the diffusion model. While an ultimate objective of the LSD construction is to develop also the generative capability, we take the perspective that the discriminative capability serves as a useful guide for identifying an appropriate autoencoder architecture, and so defer the more challenging generative aspect until this is established. This choice is grounded on the intuition that a generative model can meaningfully generate only to the extent that it can discriminate.

Through its t-dependence, the diffusion model provides a one-parameter family of learned representations. There are two subtleties to this however. The first arises from the fact that the input to the diffusion model, z_t, depends on both t and sampled ε. As ε essentially amounts to Gaussian broadening, we can treat this as regularization and employ the mean value, i.e. take

$$\bar{v}_t(z) = \texttt{Diffusion}(\cos(\pi t/2)z, t). \tag{6}$$

The second subtlety is that the diffusion model learns nothing for $t \sim 1$ as the input there is noise. This can be compensated for by switching to the score function, which from Tweedie's formula Robbins (1992) is expressed as

$$s_t(z) = -\frac{\hat{\varepsilon}_t(z)}{\sin(\pi t/2)} = -\frac{2}{\sin(\pi t)}\big(\hat{v}_t(z) + \sin(\pi t/2)z\big). \tag{7}$$

Dropping the singular prefactor, we thus take the diffusion representations as

$$\bar{v}_t(z) + \sin(\pi t/2)z. \tag{8}$$

At $t = 0$, this reduces back to $\hat{v}_t(z)$.

4 Evaluation

We present here two trained models for both LSD-TN and LSD-NM variants. We call these S and M, and provide their parameter counts in Table 1. The models are trained on the Uniref50 protein sequence dataset Suzek et al. (2015), with sequences of maximum length 254, and omitting sequences with unknown or non-canonical amino acids (0.5% of the dataset). Full model hyperparameters and further training details are given in Appendix A.

To establish a baseline for their performance we additionally train corresponding MLM models, along with a diffusion model on their learned embeddings, using an identical setup. These MLM baseline models can be viewed as counterparts to DiMA Meshchaninov et al. (2024), with self-conditioning deactivated. In terms of Fig. 1, for these MLM models the decoder's transformer trunk is trivialized. Masking is applied at the input to the encoder, as opposed the input of the decoder for LSD-NM. We employ a 15% masking rate, and extract the latent embeddings after the layer norm following the transformer layers to ensure they are appropriately normalised.

We assess the discriminative capabilities of these models across a set of a property prediction tasks assessing stability, interaction and functional characterization, which we adopt from SaProt Su et al.

Models	Thermostability ↑	HumanPPI ↑	Metal Ion Binding ↑	DeepLoc ↑	
				Subcellular	Binary
	Spearman's ρ	Acc (%)	Acc (%)	Acc (%)	Acc (%)
ESM 8M	0.648	72.7	63.2	68.2	88.8
ESM 650M	0.690	81.3	66.8	77.6	91.0
DPLM 650M	0.693	76.7	69.1	78.5	90.8
MLM-S: Encoder	0.606	72.3	65.0	60.1	86.3
MLM-S: Diffusion ($t = 0$)	0.474	57.7	63.0	46.1	74.3
MLM-M: Encoder	0.613	72.3	63.5	62.4	87.3
MLM-M: Diffusion ($t = 0$)	0.543	60.6	61.7	52.2	76.2
LSD-TN-S: Encoder	0.560	58.6	64.6	53.0	76.6
LSD-TN-S: Diffusion ($t = 0$)	0.562	62.6	62.8	48.2	75.3
LSD-TN-M: Encoder	0.571	59.1	63.2	54.4	76.7
LSD-TN-M: Diffusion ($t = 0$)	0.571	65.9	62.6	52.7	76.5
LSD-NM-S: Encoder	0.553	62.6	64.1	54.6	77.6
LSD-NM-S: Diffusion ($t = 0$)	0.567	60.2	65.0	53.5	76.1
LSD-NM-M: Encoder	0.571	61.6	64.6	55.0	77.3
LSD-NM-M: Diffusion ($t = 0$)	0.581	61.1	64.7	54.2	76.8

Table 2: Evaluation on protein property prediction tasks. The $t = 0$ diffusion representations are highlighted, green for the LSD models and gray for the MLM baseline. Reported scores are computed as the mean of 5 randomly initialized predictors.

(2023). For this we freeze the backbone and training a simple predictor on the mean of the embeddings across the sequence. We provide further information on the datasets and predictor architecture in Appendix B.

We report the performance of the models in Table 2. For the LSD-NM and LSD-TN models and the MLM diffusion baseline, we evaluate on both the latent space, at the output of the encoder, and on the $t = 0$ output of the diffusion model applied on the latent space. To further benchmark these results we additionally evaluate the predictor on two prominent protein representation learning models, ESM2 Lin et al. (2023) and the discrete diffusion model DPLM Wang et al. (2024).

We first highlight the diffusion model results, which are the primary focus of this work. We see that the LSD-TN and LSD-NM diffusion models consistently outperform the MLM diffusion models across all evaluation metrics. Comparing the between the LSD-TN and LSD-NM variants, we see that the LSD-NM performs better than LSD-TN on all but one task. The exception is the HumanPPI, on which LSD-TN-M performs notably better. This may indicate a complementarity in how the two different constructions organise correlations within their respective latent spaces.

Turning to the latent representations of the encoder we see that the situation is reversed, with the MLM results here greatly outperforming their LSD counterparts. This aligns with the well-established strength of masked language modelling for representation learning, in contrast to the LSD autoencoders which were not designed to optimise for this. Indeed, the MLM encoder performs best of all the evaluated MLM, LSD-TN and LSD-NM representations, and performs significantly better than even the best LSD diffusion models. This trend is also exhibited by the ESM 8M model, which has a smaller parameter count than all the LSD models.

We also compare between the encoder and diffusion representations. For the LSD-TN model we observe a complementarity, with results consistently better for the encoder on Metal Ion Binding and DeepLoc-Subcellular, on a par for Thermostability and DeepLoc-Binary, and better for the diffusion model on HumanPPI. For LSD-NM the results are more similar between the two modules, while for the MLM model the diffusion representations all significantly score lower than the encoder's.

We now turn to the t-dependence of the diffusion representations. In Fig. 2 we evaluate the performance of the regularised score function $\bar{v}_t(z) + \sin(\pi t/2)z$ for all five tasks for the LSD-TN-M and LSD-NM-M models. For the LSD-TN model we observe that the curves are notably flat, with the exception of HumanPPI although that could reflect the greater uncertainty in that metric. For LSD-NM on the other hand, there is some variation to the curves with different trends for the different tasks. This may reflect the expectation that different correlations are captured at the different scales